B. Different String

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: implementation, strings | https://codeforces.com/problemset/problem/1971/B

You are given a string s consisting of lowercase English letters.

Rearrange the characters of s to form a new string r that is **not equal** to s, or report that it's impossible.

Input

The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

The only line of each test case contains a string s of length at most 10 consisting of lowercase English letters.

Output

For each test case, if no such string r exists as described in the statement, output "NO" (without quotes).

Otherwise, output "YES" (without quotes). Then, output one line — the string r, consisting of letters of string s.

You can output "YES" and "NO" in any case (for example, strings "yEs", "yes", and "Yes" will be recognized as a positive response).

If multiple answers are possible, you can output any of them.

Example

Input
8 codeforces aaaaa xxxxy co d nutdealer mwistht hhhhhhhhhh

Output
YES forcodesec NO YES xxyxx YES oc NO YES undertale YES thtsiwm NO

Note

In the first test case, another possible answer is `forcescode`.

In the second test case, all rearrangements of aaaaa are equal to aaaaa.